

Time-series Foundation Model Evaluation

Experiment Guideline

Thesis project

25 August 2026

1 Scope

This is an inference-only evaluation of frozen Chronos-2 and Chronos-Bolt, official TS-ICL and TiRex-2 inference packages, in-context TabPFN-TS, and four legacy TimeTensor baselines: persistence, expected value, repeat, and weekly aligned lookback. It reproduces and extends the accepted ICLR TSALM covariate study. It contains no optimization, forecasting-model training, PatchTST training, or data-generating process.

2 Forecasting and evaluation contract

Targets are wide univariate panels with shape (users, 1, dates). For query date t ,

$$X = (t - L, t], \quad Y = (t, t + H].$$

Query dates are deterministic, date-major, spaced by `evaluation.stride` (512 for the paper protocol), and evaluated for every user. Evaluation fractions own target dates. Pairwise constant-window removal removes only user/query pairs whose lookback standard deviation is at most the configured threshold.

Target-only reversible instance normalization standardizes the target history before inference and reverses it on predictions. Covariates remain on their raw source scale. Evaluation nMSE always divides errors by the lookback scale, independently of preprocessing.

3 Models and covariates

Axis	Cases
Models	Persistence, expected value, repeat, weekly aligned lookback, frozen Chronos-2 and Chronos-Bolt, official TS-ICL and TiRex-2 adapters, and in-context TabPFN-TS.
Covariates	none ; identity (target history supplied as known covariate); known (prepared target-aligned covariate panels).
Controls	Instance normalization on/off; constant-window removal on/off; TabPFN time features on/off where applicable.

External covariate panels must share the target timeline and either match user names/order or contain one global column.

The lookback baseline infers the number of observations in seven days from dated inputs, or accepts an explicit positive period override. For horizon H and period P , it returns the latest complete H -step history block whose start is shifted back by $\lceil H/P \rceil P$ observations. This preserves weekly alignment and prevents overlap with the forecast horizon. Inputs without a cadence that divides one week require an explicit period. The former absolute-index variants are not part of the current model set.

All foundation-model adapters use their official packages. TS-ICL and TiRex-2 retain native preprocessing and test-time behavior and select the median point forecast. The TSFM boundary supplies raw targets for these two models, so the foundation workflow disables the outer instance-normalization wrapper. Their sole shared aliases are `chronos2`, `chronos_bolt`, `ts_icl`, `tirex2`, and `tabpfn_ts`. Covariate configuration does not alter model identity, and an adapter without covariate support raises on a non-empty covariate input. Expected checkpoints are

- Chronos-2: `weights/chronos2`;
- Chronos-Bolt: `weights/chronos-bolt-base`;
- TS-ICL: `weights/tsicl/tsicl-v1.ckpt`;
- TiRex-2: `weights/tirex2/model-config.yaml` and `weights/tirex2/model.ckpt`;
- TabPFN-TS: `weights/tabpfnts/tabpfnt-v2.5-regressor-v2.5_default.ckpt`.

4 Metrics and seed

Every run reports MSE, MAE, nMSE, nMAE, and MASE with configurable seasonality. For each metric it records the sample mean and population standard deviation, equal-user mean and population standard deviation across user means, the mean of the worst $\lceil 0.1U \rceil$ users, per-user within-user dispersion, and per-horizon mean and dispersion. It also records total model inference time and amortized time per user and per series/window; model and data loading are excluded from inference timing. One run seed fixes every stochastic choice exposed by a model or sampler; deterministic evaluation ordering remains fixed.

5 Experiment families and grid

The root fronts are:

1. `01_univariate.slurm`: the four baselines and Chronos-2;
2. `02_controls.slurm`: normalization and constant-window controls;
3. `03_covariates.slurm`: univariate and identity-covariate inference, with prepared known covariates available by override;
4. `04_foundation_models.slurm`: frozen Chronos-2, Chronos-Bolt, TS-ICL, and TabPFN-TS with model-native preprocessing. TiRex-2 remains implemented but is excluded from the launcher until explicitly restored.

`test` is Electricity at 504:168, seed 1. `full` crosses Electricity, Traffic, Solar, and Exchange Rate with 168:24, 336:48, and 504:168. `ultra` adds the documented extended datasets at the shared 336:48 and 504:168 settings. Supersets select from the same identity trees and reuse only signature-matched completed runs.

The foundation-model family uses Electricity at 504:168 for `test`; its `full` profile crosses Electricity, Traffic, Solar, Weather, and Exchange Rate with 336:48 and 504:168. Its report fixes Chronos-2 as the default reference. `ultra` adds ETTh1 and 168:24.

6 Execution

```
EXPERIMENT_MODE=test sbatch 01_univariate.slurm
EXPERIMENT_MODE=full sbatch 01_univariate.slurm
EXPERIMENT_MODE=full sbatch 02_controls.slurm
EXPERIMENT_MODE=full sbatch 03_covariates.slurm
EXPERIMENT_MODE=test sbatch 04_foundation_models.slurm
EXPERIMENT_MODE=full sbatch 04_foundation_models.slurm
```

Each command has a matching `_selena.slurm` overflow front. It executes the same scientific implementation on Selena but writes Slurm streams under `logs_selena/` and all manifests, results, and reports under `outputs_selena/`. DGX fronts retain `logs/` and `outputs/`. The default workflow is `STAGES=evaluate,report`; stage selection is a recovery override. Dataset, setting, model, seed, normalization, constant handling, covariates, and time-feature controls can be narrowed with the documented environment overrides. Foundation-model inference is remote only. `LOOKBACK_PERIOD_STEPS` overrides weekly cadence inference for the lookback baseline, and `MASE_SEASONALITY` sets the MASE denominator lag.

Slurm workflows never submit a publisher or execute Git commands. After any terminal state, including failure, cancellation, or timeout, the user runs `bash publish_job.sh <job-id>` from the project root. The script verifies `main`, sources `$HOME/codes/proxy.sh` (or `PROXY_SCRIPT_PATH`), and fast-forward pulls `origin/main` before artifact selection or staging. With a job ID, it commits only that job's exact stdout/stderr pair. Omitting the job ID publishes all logs and lightweight outputs, excluding `*.pt`, `*.npz`, and `*.cbm`. It then pushes `origin/main` without opening a pull request. A non-fast-forward pull stops without a merge commit, and unrelated staged paths remain outside the publisher commit.

7 Data provenance and artifacts

The loader discovers `config.json` beside the CSV/tensor directory. Portable settings are top-level, project overrides live under `tsfm_evaluation`, explicit run values override both, and `drop_users` is additive. The selected path and applied keys are logged. Tensor directories follow the current TimeTensors contract: `values.pt`, optional `datetimes.pt` and `individual_ids.pt`, optional `individual_context.pt` and `global_context.pt`, and `dataset_metadata.json`. Static individual context is broadcast across dates and global context across users.

Selective TIME preparation

`python -m scripts.prepare_time_csv` resolves an immutable revision of Real-TSF/TIME-ProcessedCSV and downloads only selected CSV paths. The default considers 15-minute, hourly, and daily sources, retaining source tasks with at most 500 usable series and at most 10,000 dates per series. Files are merged only when their complete timestamp indices match; non-aligned groups become separate datasets, while irregular, short, or non-finite series are recorded as skips. Each accepted panel is written below `datasets/time/<dataset>/` with a directly loadable wide CSV and `config.json`. `datasets/time/catalog.json` records source revision, counts, skips, and eligible query/window totals for the requested settings and stride. The current revision inventory retains five hourly and seven daily source configurations under the default limits; no 15-minute source qualifies.

The four independent family roots live below `outputs/` on DGX and below `outputs_selena/` on Selena. Their relative names are `univariate`, `controls`, `covariates`, and `foundation_models`. Their ordered identity is

```
outputs/family/dataset/L_H/backbone/covariate_mode/normalization/  
    constant_policy/time_features/run_n/
```

Every model config owns one directory. Batch size, data split and stride, metric/window export, model inference options, and explicit covariate bindings are pipeline configs; device and Slurm placement are runtime-only. One seed fixes every stochastic choice. Mode filters path subsets and is not part of the path or computation signature. TSFM launches one seed per run, records it in the manifest, and stores its artifacts directly under `run_n`; a seed leaf is reserved for a future multi-seed launcher. The manifest structure, statuses, collision and selection policies, purposes, and report manifests are thesis-wide conventions. The workflow roots, ordered identity, and required summary artifacts are TSFM-specific.

Each completed run contains `summary.json`, `per_user_metrics.csv`, `horizon_metrics.csv`, optional `window_metrics.csv`, `resolved_config.json`, and a `schema_version: 1` manifest with declared configs, optional plain provenance, launch attempts, required artifacts, per-seed state, and one of four overall statuses; seed state may additionally be `ready`. Run identity contains only the schema, identity/model configs, pipeline/experiment configs, and seeds. It never fingerprints or hashes code, Slurm files, data, weights, logs, outputs, or directories; code/data changes are manual rerun decisions. The schema changes only for a deliberate global contract break, and `new` creates another repeat with unchanged parameters. The default `overwrite_exact` policy skips exact completed work, resumes exact interruption, and creates a new `run_n` for a changed pipeline. The evaluator records ready artifacts, and completion is written immediately after that evaluation's `srun` returns successfully. Later evaluation or report failures preserve completed runs and interrupt only unfinished work. Completed manifests are authoritative, without later file hashing or synchronization checks. Reports accept distinct/latest/average config handling, selected/latest/distinct/average repeat selection, explicit pipeline filters, and purpose filters. Explicit pipeline filters select configurations and must match even with one run. Nested pipeline and experiment fields, including the scientific identity embedded for an upstream dependency, are exposed through dotted filter keys and every differing nested field appears in a distinct label. If latest-run timestamps tie, the larger `run_n` index wins, making selection deterministic. Reports write requested filters and obtained inputs to `outputs/reports/family/mode` on DGX or the same relative path below `outputs_selena/` on Selena, while `SELECTED_RUNS.txt` stores only automatic or pinned exact repeats.

Reports compare complete configurations without mixing incompatible pipeline settings. They write the selected raw results, model-versus-reference configuration table, equal-configuration marginal tables by dataset and by setting, Chronos-2 strict paired-window win rates against the best aggregate baseline, and a report manifest. The selected table metric may be any error metric or total/per-user/per-window inference time. When window artifacts are enabled, reports also export per-user mean-versus-dispersion scatters, window error histograms, and horizon-error curves as PNG and PDF with a plot index.

8 Validation

The shared scientific runtime may run dataset, metric, model-adapter, repeat/report, manifest, and Slurm workflow checks. Local checks may use fake adapters or the repeat baseline, but must not load Chronos-2, Chronos-Bolt, TS-ICL, TiRex-2, or TabPFN-TS checkpoints or run full sweeps.